greatest mathematical erudition—he may have more mathematical culture than anyone in modern times, except perhaps for his late father, Adrien Douady.

At the time of writing, we may have produced a formal proof using mathematics, and a branch of mathematics called "measure theory" that was used by the French to put rigor behind the mathematics of probability. The paper is provisionally called *"Undecidability: On the inconsistency of estimating probabilities from a sample without binding a priori assumptions on the class of acceptable probabilities."*

It's the Consequences …

Further, we in real life do not care about simple, raw probability (whether an event happens or does not happen); we worry about consequences (the size of the event; how much total destruction of lives or wealth, or other losses, will come from it; how much benefit a beneficial event will bring). Given that the less frequent the event, the more severe the consequences (just consider that the hundred-year flood is more severe, and less frequent, than the ten-year flood; the bestseller of the decade ships more copies than the bestseller of the year), our estimation of the *contribution* of the rare event is going to be massively faulty (contribution is probability times effect; multiply that by estimation error); and nothing can remedy it.[*]

So the rarer the event, the less we know about its role—and the more we need to compensate for that deficiency with an extrapolative, generalizing theory. It will lack in rigor in proportion to claims about the rarity of the event. Hence theoretical and model error are more consequential in the tails; and, the good news, *some representations are more fragile than others*.

I showed that this error is more severe in Extremistan, where rare events are more consequential, because of a lack of scale, or a lack of asymptotic ceiling for the random variable. In Mediocristan, by comparison, the collective effect of regular events dominates and the exceptions are rather inconsequential—we know their effect, and it is very mild because one can diversify thanks to the "law of large numbers." Let me provide once again an illustration of Extremistan. Less than 0.25 percent of all the companies listed in the world represent around half the market capitalization, a less than minuscule percentage of novels on the planet accounts for approximately half of fiction sales, less than 0.1 percent of drugs generate a little more than half the pharmaceutical industry's sales—and less than 0.1 percent of risky